

Data-Transfer Group

1) MVI B, 25H

$B \leftarrow 25H$

2) LXI B, 2000H

$B, C \leftarrow 2000H$
(20)(00)

3) MVI M, 25H

$M \leftarrow 25H$

4) MOV B, C

$B \leftarrow C$

5) MOV B, M

$B \leftarrow M$

6) MOV M, C

$M \leftarrow C$

7) LDA 2000H

8) STA 2000H

9) LHLD 5000H

10) SHLD 2000H

11) LDAX B

12) STAX B

13) PCHL ($PC \leftarrow HL$)

14) SPHL ($SP \leftarrow HL$)

15) XCHG ($HL \leftrightarrow DE$)

16) XTHL ($[SP] \leftrightarrow HL$)

Respect



Edits

Data Xster Group.

1) MVI R, data
Eg: MVI B, 25H

2) LXI Rp, Data
Eg: LXI B, 2000H

3) MVI M, data
Eg: MVI M, 25H

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4) MOV Rd, Rs
Eg: MOV B, C

5) MOV B, m

6) MOV m, C

7) LDA

8) S

9) LDAX B

13) PCHL

14) SPHL

15) XCHG

16) XTHL

MOV, MVI, LXI, LDA, STA, LDAX, STAX, LHLD, SHLD, PCHL, SPHL, XCHG, XTHL



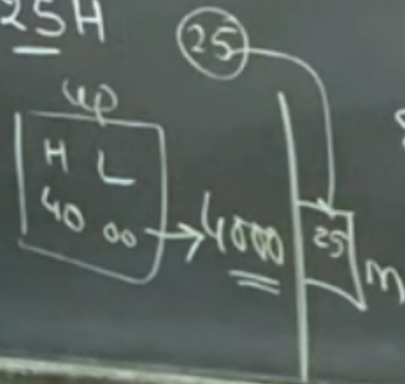
1) MVI R, data
Eg: MVI B, 25H
B ← 25H

2) LXI Rp, Data
Eg: LXI B, 2000H
BC ← 2000
B C

3) MVI M, data

Eg: MVI M, 25H

M ← 25H
[HL] ← 25H



5) MOV B, m

6) MOV m, C

7) LDA 2000

8) STA 3000

1] Data Xster Group.

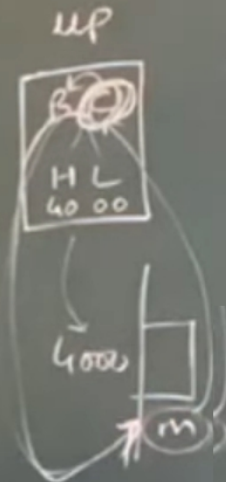
1) movl R, data
B, 25H
B ← 25H

Data

B, 2000H

B ← 2000
B C

lata



4) mov Rd, RS
Eg: mov B, C

B ← C
25 ← 25

5) mov B, m
B ← m

6) mov m, C
m ← C

7) LDA 2000H
A ← [2000H]

8) STA 3000H
A → [3000]
2000 | 33
3000 | 25

a) LDA

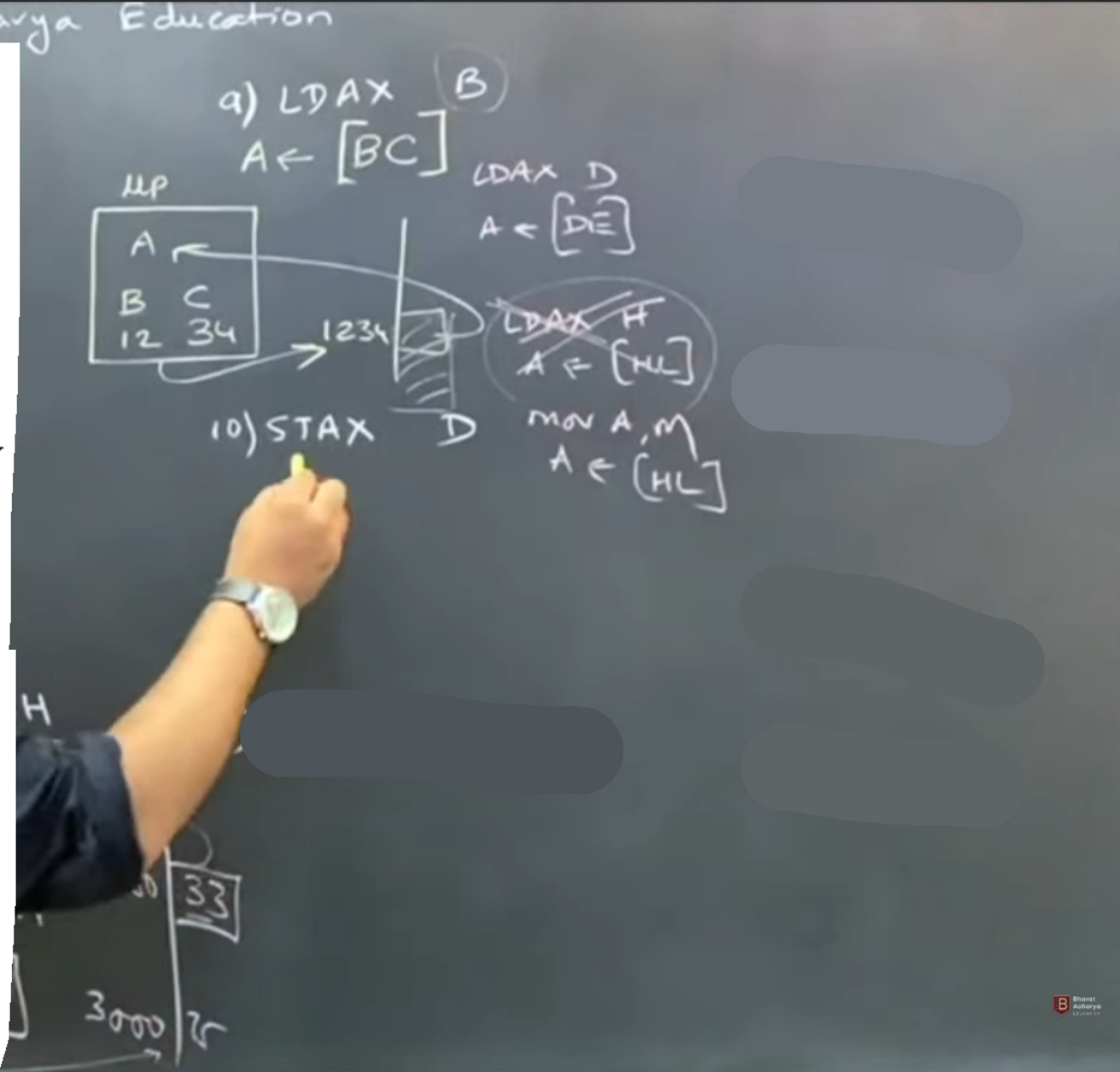
10) STAX

11) LHLD

LDAX Rp & STAX Rp
works on 8-bit data

- Load Accum. with the data pointed by Rp memory location.

- LDAX & STAX are indirect addressing instructions.



LHLD & SHLD works with 16-bit Data

Eg: `mov B, C`

10) `STAX` → `mov A, m`
`A ← [HL]`

MP `A → [DE]` `A → [BC]`
`mov m, A`
`A → m`

Diagram for `STAX`:
A = 75
DE = 4000
Memory location 4000 contains 75.

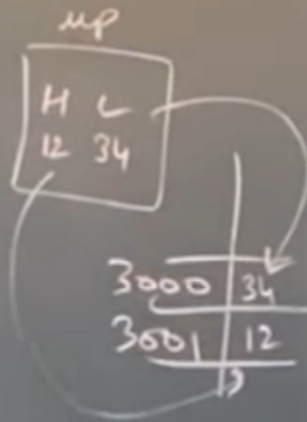
11) `LHLD 2000H`
`L ← [2000]`
`H ← [2001]`

Diagram for `LHLD`:
Memory location 2000 contains 33.
Memory location 2001 contains 12.
HL = 1234.

12) SHLD 3000H

L $\rightarrow$ [3000]

H $\rightarrow$ [3001]



13) PCHL

PC $\leftarrow$ HL

14) SPHL

SP $\leftarrow$ HL

15) XCHG HL $\leftrightarrow$ DE

II Arithmetic Group

1) ADD B

7) ADC B

2) ADD m

8) ADC m

9) ACI 25H

10) SBB B

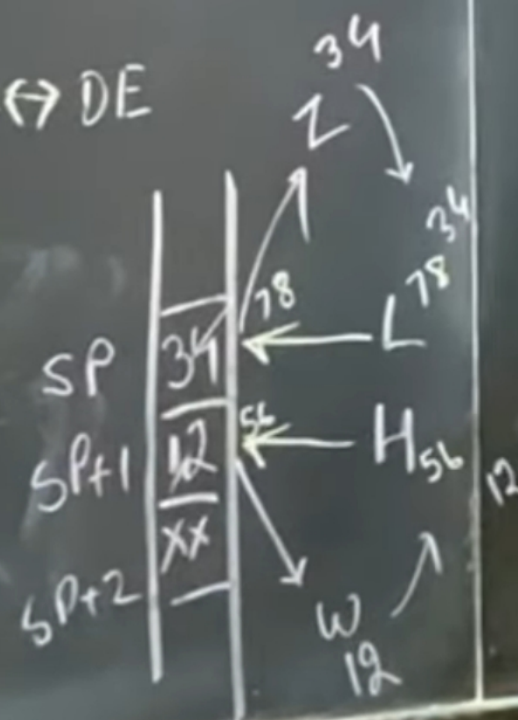
8085
Bharat

HL

14) SPHL
 $SP \leftarrow HL$

15) XCHG HL $\leftrightarrow$ DE

~~16) XTHL~~
 $L \leftrightarrow [SP]$
 $H \leftrightarrow [SP+1]$
 (w)
1234



2) ADD m

3) ADI 25H

4) SUB B

5) SUB

6)

14) INX

15) INR

16) DCR

17) DCX